








Time Series Analysis

Seminar 13: Speculative bubbles: LPPL models



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Acronyms used in this material

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **BET**: Bucharest Exchange Trading (index)
- **BSADF**: Backward Supremum Augmented Dickey–Fuller (statistic)
- **EODHD**: EOD Historical Data (market-data provider)
- **GSADF**: Generalised Supremum Augmented Dickey–Fuller (test)
- **LPPL**: Log-Periodic Power Law
- **OLS**: Ordinary Least Squares
- **PSY**: Phillips–Shi–Yu (tests)
- **SADF**: Supremum Augmented Dickey–Fuller (test)

Today's Question and Seminar Structure

- The question of the seminar
  - **can we see a bubble while it grows, date it, and say when it ends?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C1
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Chapter 13 is a self-study chapter: read it before the seminar; the recap gives the formulas used today
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the right-tailed ADF on one window; reading SADF and GSADF	Solved, Proposed	A1
A3	dating a bubble from the BSADF sequence	Solved	A3
A4	the LPPL formula and its filter conditions	Proposed	A3
A5	the recursion of a rational bubble	Solved	A5
B1, B2	the S&P 500 in the dot-com years; LPPL fits of Bitcoin in 2021	Solved, Proposed	B1
C1	what is wrong in an AI answer?	Proposed	A1–B2

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
S&P 500	EODHD (GSPC.INDX)	weekly, Friday close	1990–2004
Bitcoin	EODHD (BTC-USD.CC)	daily, 7 days a week	2020–2021
Nasdaq 100, Bitcoin, BET, Shanghai	Chapter 13 results	weekly	1990–2023

- Weekly log price $y_t = \ln P_t$ for the explosive-root tests; daily log price for the LPPL fits; time in decimal years
- In the notebook: `adf_window`, `psy`, `psy_cv`, `episodes_above`, `lppl_fit`, `lppl_conditions` (the functions of Chapter 13); no account or key is needed

Recap (1/3): explosive roots and the PSY tests

- **Right-tailed ADF** on a window $[r_1, r_2]$ of the sample (Augmented Dickey–Fuller, no lagged differences here) Phillips, Wu and Yu (2011)

$$\Delta y_t = a + \delta\, y_{t-1} + \varepsilon_t, \qquad H_0 : \delta = 0 \text{ (unit root)}, \quad H_1 : \delta > 0 \text{ (explosive, } \rho = 1 + \delta > 1)$$

  - statistic $\mathrm{ADF}_{r_1}^{r_2} = \hat{\delta}/\mathrm{se}(\hat{\delta})$; reject H_0 when it is **above** the right-tail critical value

- **Sup statistics** (PSY: Phillips, Shi and Yu) Phillips, Shi and Yu (2015a); smallest window $r_0 = 0.01 + 1.8/\sqrt{T}$

$$\mathrm{SADF} = \sup_{r_2 \in [r_0, 1]} \mathrm{ADF}_0^{r_2}, \qquad \mathrm{GSADF} = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}, \qquad \mathrm{BSADF}_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$$

  - critical values by Monte Carlo under a random walk with a weak drift (1000 replications in the course)

- **Date-stamping**: an episode starts at the first week with BSADF_{r_2} above its 95% critical value and must last at least $L = [\ln T]$ weeks (rounded)

Recap (2/3): rational bubbles

- **Price = fundamental value + bubble**: with a required return r, $p_t = p_t^* + b_t$ and Blanchard and Watson (1982)

$$\mathbb{E}_t(b_{t+1}) = (1 + r)\, b_t$$

  - constant dividend D: $p^* = D/r$; a bubble is the part of the price that dividends do not explain
- **Blanchard–Watson**: each period the bubble survives with probability π, otherwise it bursts

$$b_{t+1} = \begin{cases} \dfrac{1+r}{\pi}\, b_t + \varepsilon_{t+1} & \text{with probability } \pi \\ \varepsilon_{t+1} & \text{with probability } 1 - \pi \end{cases}$$

  - while it survives the bubble grows faster than r: the extra growth pays for the risk of the crash
  - the duration is geometric: mean $1/(1-\pi)$ periods (as the regimes of Chapter 10)

Recap (3/3): the LPPL model and its filter

- **LPPL** (log-periodic power law) Johansen, Ledoit and Sornette (2000): the log price before a critical time t_c

$$\ln p(t) = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos\left(\omega \ln(t_c - t) - \phi\right), \qquad t < t_c$$

  - $B < 0$, $0 < m < 1$: faster than exponential growth; ω: frequency of the oscillations in $\ln(t_c - t)$; successive peaks shrink by $\lambda = e^{2\pi/\omega}$
  - Filimonov–Sornette: A, B, C_1, C_2 by OLS (ordinary least squares) for given (t_c, m, ω) Filimonov and Sornette (2013)
- **Filter** of a fit on $[t_1, t_2]$ Shu and Zhu (2020): $B < 0$; $0.01 \le m \le 0.99$; $2 \le \omega \le 25$; $t_2 \le t_c \le t_2 + (t_2 - t_1)/5$

$$\text{oscillations} = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \ge 2.5, \qquad \text{damping} = \frac{m\,|B|}{\omega\,|C|} \ge 1, \qquad \text{relative error} \le 0.15$$

  - $C = \sqrt{C_1^2 + C_2^2}$; damping ≥ 1 keeps the fitted price increasing; t_c is the most likely end of the regime, not a crash date

A1: the right-tailed ADF on one window [Solved]

Task

- Weekly S&P 500 log prices, 5 January 1990 to 24 March 2000 (the peak week): $n = 533$ weekly changes. OLS gives $\hat{a} = -0.0084$, $\hat{\delta} = 0.00175$, $\text{se}(\hat{\delta}) = 0.00180$.
- 1. Compute the ADF statistic and the root $\hat{\rho}$.
- 2. The 95% critical value of BSADF for this week is 0.542. Is the week explosive?
- 3. The weeks above the critical value around the peak run from 3 March 2000 to 7 April 2000; with $L = 7$, is this a dated episode?
- Report: one statistic, one root, two decisions.

Solution

- 1. ADF $= 0.00175/0.00180 = 0.970$; $\hat{\rho} = 1 + \hat{\delta} = 1.00175$
- 2. $0.970 > 0.542$: reject the unit root in favour of an explosive root (right tail); a left-tailed reading ($t < -2.86$) would wrongly accept
- $\hat{\rho}$ is barely above 1: a deviation doubles in $\ln 2 / \ln \hat{\rho} \approx 398$ weeks; mild explosiveness is enough to be detected
- 3. No: the run lasts 6 weeks, fewer than $L = 7$; the minimum duration removes short crossings

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A2: the data [Proposed]

Series (weekly)	T	w_0	SADF	cv_{95}	GSADF	cv_{95}	Episodes
Nasdaq 100, 1990–2004	782	?	2.722	1.481	2.925	2.276	6
Bitcoin, 2014–2023	484	?	3.849	1.495	3.849	2.184	2
BET, 2000–2012	663	52	1.859	1.485	3.533	2.294	3
Shanghai, 2010–2018	460	43	1.501	1.406	3.458	2.147	1

- Results of Chapter 13 (psy, psy_cv, 1000 replications); cv_{95}: 95% critical value; w_0: smallest window in weeks
- Nasdaq 100, whole-sample ADF: -1.272, with the 95% critical value -0.132

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A2: reading SADF and GSADF [Proposed]

Solution discussed in class

Task

- The table of the previous slide. Model: A1.
- 1. Compute $r_0 = 0.01 + 1.8/\sqrt{T}$ and $w_0 = \lfloor r_0 T \rfloor$ for the Nasdaq 100 and for Bitcoin.
- 2. For each series, decide on SADF and on GSADF at 5%.
- 3. Why is GSADF never below SADF? Why are they equal for Bitcoin?
- 4. Why does the whole-sample ADF of the Nasdaq 100 not reject?
- Report: two windows, eight decisions, two sentences.

A3: the data [Solved]

Week	BSADF	cv_{95}
26 May 1995	0.188	0.484
2 June 1995	0.518	0.427
9 June 1995	0.219	0.436
16 June 1995	0.657	0.443
23 June 1995	1.022	0.444
30 June 1995	0.668	0.473

Week	BSADF	cv_{95}
7 July 1995	1.090	0.461
14 July 1995	1.149	0.419
21 July 1995	0.745	0.475
28 July 1995	1.046	0.475
4 August 1995	0.783	0.475
11 August 1995	0.563	0.433

- Weekly S&P 500, 1990–2004 ($T = 782$ weekly changes, B1); BSADF and its 95% critical value, weeks around the first crossing that lasts

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A3: dating a bubble from BSADF [Solved]

Task

- The table of the previous slide; the episode rule of the recap.
- 1. Compute the minimum duration $L = [\ln T]$.
- 2. Mark the weeks with BSADF above the critical value.
- 3. Give the start of the first episode. Why is the first crossing not the start?
- Report: L, a start date, one sentence.

Solution

- 1. $\ln 782 = 6.66$, so $L = 7$ weeks
- 2. Above: 2 June 1995 ($0.518 > 0.427$), then every week from 16 June 1995 on
- 3. The episode starts on 16 June 1995 and lasts 57 weeks (to 12 July 1996); the crossing of 2 June 1995 lasts one week, fewer than L
- Over the whole sample, 8 of the 13 crossings are shorter than L and are dropped

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A4: the LPPL formula and its filter [Proposed]

Solution discussed in class

Task

- LPPL with $A = 8$, $B = -1$, $C = 0.1$, $m = 0.4$, $\omega = 7$, $\phi = 0$, $t_c = 1$ (time in years). Model: A3.
- 1. Compute $\ln p(t)$ at $t = 0,\ 0.5,\ 0.9$, term by term.
- 2. Compute $\lambda = e^{2\pi/\omega}$ and the first three peaks of the cosine after $t = 0$.
- 3. For a window $[t_1, t_2] = [0, 0.9]$, compute the oscillations and the damping. Which filter condition fails?
- Report: three log prices, λ, three dates, two numbers, one verdict.

A5: the recursion of a rational bubble [Solved]

Task

- Blanchard–Watson bubble with $r = 0.01$ per period, $\pi = 0.95$, $b_0 = 1$, no noise; a constant dividend $D = 2$.
- 1. Compute the growth factor while the bubble survives and the path b_1, b_2, b_3.
- 2. Check that $\mathbb{E}_0(b_1) = (1 + r)\, b_0$.
- 3. Compute the probability of surviving three periods, the mean duration and the median duration.
- 4. Compute p^* and the share of the bubble in the price after three periods.
- Report: one factor, three values, three probabilities or durations, one share.

Solution

- 1. $(1 + r)/\pi = 1.01/0.95 = 1.0632$ (+6.32% per period); $b_1 = 1.0632$, $b_2 = 1.1303$, $b_3 = 1.2017$
- 2. $\mathbb{E}_0(b_1) = \pi \cdot 1.0632 \cdot 1 + (1 - \pi) \cdot 0 = 1.01 = (1 + r)\, b_0$
- 3. $\pi^3 = 0.857$; mean $1/(1 - \pi) = 20$ periods; median $\ln 0.5 / \ln \pi = 13.5$ periods
- 4. $p^* = 2/0.01 = 200$; $p_3 = 201.20$, bubble share 0.60%: a young bubble is invisible in the price; it matters only after many periods of survival

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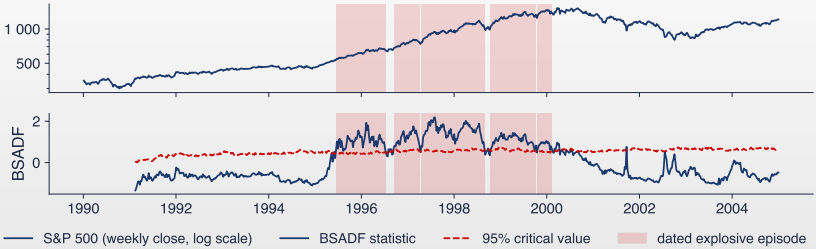
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B1: the S&P 500 in the dot-com years [Solved]

- **Question**: was the S&P 500 explosive in the late 1990s, and when, compared with the Nasdaq 100 of Chapter 13?
- **Inputs**: weekly S&P 500 log prices, January 1990 to December 2004 ($T = 782$), the same sample as the Nasdaq 100 of the lecture
- **Tasks**
  1. Compute ADF, SADF, GSADF and the BSADF sequence (`psy`) and their critical values (`psy_cv`, 1000 replications).
  2. Date the episodes with the 95% critical value of BSADF and $L = [\ln T]$ (`episodes_above`).
  3. Compare the statistics and the longest episode with those of the Nasdaq 100.
  4. Interpretation: how strong is the evidence of a bubble in the broad market?
- **Report**: three statistics with their critical values, a list of episodes, one sentence
- Notebook: section B1

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B1: solution [Solved]



- ADF -1.087 (cv_{95} -0.132); SADF 1.804 (cv_{95} 1.481); GSADF 2.194 (cv_{90} 2.028, cv_{95} 2.276); $w_0 = 58$ weeks, $L = 7$
- 5 episodes, 228 weeks in total, from 16 June 1995 to 11 February 2000; the longest: 18 April 1997 to 28 August 1998 (72 weeks). Nasdaq 100: SADF 2.722, GSADF 2.925; longest episode 23 October 1998 to 3 November 2000 (107 weeks)
- Interpretation: weaker than for the Nasdaq: SADF rejects at 5%, GSADF only at 10%, so the dated episodes are a description, not a 5% test result; the explosive phase of the S&P 500 ends before the peak of March 2000, that of the Nasdaq goes on to November 2000

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B2: LPPL fits of Bitcoin in 2021 [Proposed]

- **Question**: would an LPPL fit have announced the Bitcoin peak of April 2021, and how does the estimated critical time move as the window grows?
- **Inputs**: daily Bitcoin log price from the low of 8 September 2020 (10132 USD) to the peak of 13 April 2021 (63503 USD); model: A4 and Chapter 13
- **Tasks**
  1. Fit the LPPL model (`fit_window`) on windows that start at the low and end 90, 60 and 30 days before the peak.
  2. For each fit report t_c, m, ω and the distance between t_c and the actual peak.
  3. Apply the filter (`lppl_conditions`) and say which conditions fail.
  4. Interpretation: is t_c a forecast of the crash date?
- **Report**: three critical times, six parameters, three lists of failed conditions, one sentence
- Notebook: section B2

C1: audit an AI answer [Proposed]

- A student asked an AI assistant about bubble tests and LPPL models. The answer:
- (a) To test for a bubble, run the ADF test and reject the unit root when the statistic is below -2.86.
- (b) The SADF of the Nasdaq 100 is above its critical value, so the Nasdaq was in a bubble from 1990 to 2004.
- (c) GSADF can be smaller than SADF when the bubble is at the end of the sample.
- (d) The LPPL critical time tc is the date of the crash: sell the day before tc.
- (e) A fit with m = 0.999 and omega = 1 is a valid bubble signal if its errors are small.
- (f) In a rational bubble the expected bubble grows at the rate r, although it grows faster than r while it survives.
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (Parts A and B, notebook section C1).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- Bubble tests are right-tailed: $H_1 : \rho > 1$; a whole-sample ADF misses a bubble that collapses inside the sample
- SADF and GSADF answer "is there an episode?"; BSADF with the minimum duration $[\ln T]$ answers "when?"
- The S&P 500 shows weaker evidence than the Nasdaq 100 in the same years
- LPPL: check the filter before reading t_c; t_c moves with the window and is not a crash date
- A rational bubble grows by $(1 + r)/\pi$ while it survives and by $1 + r$ on average

After the seminar

- Chapter 13 (self-study) develops each topic of today
  - the episodes, the PSY tests, the LPPL calibration, the confidence indicator and its evaluation
- Try the [Proposed] tasks in the notebook
- A project idea: date-stamp the BET and the Bucharest sector indices with PSY and compare with the LPPL confidence indicator
- Reading: Phillips, Shi and Yu (2015a); Johansen, Ledoit and Sornette (2000); Filimonov and Sornette (2013); Shu and Zhu (2020)

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